

CALCULUS 101 · WEEK 2 QUIZ

Calculus 101 — Week 2

Name: _____

Date: _____

How to play: Read each question twice, read every choice, cross out the wrong ones, then put an X in the box next to your answer. Get **6 of 8** right to earn the badge.

1. A feeding pump runs at 4 mL per hour for 24 hours. How much milk did the baby get?

☐

A. 28 mL

☐

B. 6 mL

☐

C. 96 mL

☐

D. 48 mL

2. On a graph with the pump rate as height and hours along the bottom, the total amount given is...

☐

A. the steepness of the rate line at any moment.

☐

B. the area of the shape under the rate line.

☐

C. the height of the rate line at the end of the day.

☐

D. the length of the rate line from start to finish.

3. The pump runs at 2 mL/hr for 6 hours, then 4 mL/hr for 18 hours. What's the day's total?

☐

A. 48 mL

☐

B. 24 mL

☐

C. 144 mL

☐

D. 84 mL

4. A baby gets a 15 mL feed every 3 hours. How much in a full day?

☐

A. 360 mL

☐

B. 120 mL

☐

C. 15 mL

☐

D. 45 mL

5. A 1-kilogram preemie takes 150 mL of milk in a day. In the doctors' units, that is...

- ☐ A. 150 mL/kg/day — about 'full feeds'
- ☐ B. 15 mL/kg/day — a trophic amount
- ☐ C. 150 mL per hour — a fast drip
- ☐ D. 1,500 mL/kg/day — far too much

6. When the rate is a smooth curve instead of flat or stepped, how do you find the total?

- ☐ A. Use the rate at the very end of the curve and multiply it by the total time.
- ☐ B. Slice time into thin strips, treat each as a skinny rectangle, and add them all up.
- ☐ C. Measure how steep the curve is in the middle and multiply that by the height.
- ☐ D. Use the highest rate anywhere on the curve and multiply it by the total time.

7. Why do NICU nurses weigh diapers?

- ☐ A. The weight tells them how much milk the baby drank at the last feed.
- ☐ B. A heavier diaper means the baby is gaining weight, which counts toward growth.
- ☐ C. Diapers are weighed to check that the right size was put on the baby.
- ☐ D. Each extra gram is a milliliter of urine, so the diapers add up to the day's output total.

8. How did Archimedes find the area of a circle, around 250 BC?

- ☐ A. He filled a circular bowl with water and weighed the water on a balance.
- ☐ B. He measured around the edge of a circle with a rope and divided by three.
- ☐ C. He squeezed it between straight-sided shapes with more and more sides until the gap vanished.
- ☐ D. He cut a circle out of bronze and compared its weight to a square of bronze.

Score: _____ / 8

Badge earned? YES / NOT YET (need 6)